Reduced coordinates for the Collatz map: exact per-step laws, anchor dynamics, and the limits of counting arguments for cycles

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Abstract

We study the Collatz dynamics in a reduced coordinate system that compresses each deterministic valuation run into a single block, producing a self-map F on states (ω, d) — an odd core prime to 3 and a depth. The reduction is faithful: the Collatz conjecture is equivalent to every F-orbit reaching $(1, 1)$, with nontrivial cycles in bijection. In these coordinates the local arithmetic admits exact laws. A single 2-adic quantity, the *anchor* $M(\omega) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, governs the step: the exit valuation obeys the global law $s = 2 + v_2(d - M(\omega))$ whenever $3^d\omega \equiv 1 \pmod 8$ and is constant on the remaining residue classes; the depth evolution closes exactly in terms of the anchor displacement together with a stated 3-adic absorption law; and the anchor increment along one step obeys an exact law modulo any power of 2, computable from graded residues of the state. A finite window of digits, together with the step's stratum labels, consequently decides each step in an error-free trichotomy, while a digit-budget accounting indicates that no bounded window can decide infinite horizons — localizing the difficulty of the problem in the digit supply of the anchors. On the cycle side, a one-line elimination identity yields short rederivations of the classical exclusions for cycles of one, two, and three blocks. Our main new theorem is a sharp dichotomy for counting arguments: a trim uniform in the number of blocks p exists, giving effective finiteness at every period, but its constant necessarily degrades like $(\log_2 3)^{-p}$ — an explicit family of near-counterexamples (*staircases*: geometric climbs closed by a single crash, precisely divergent-orbit profiles bent into loops) shows counting arguments cannot do substantially better, so uniform cycle exclusion requires arithmetic (divisibility) input, not sharper counting. Finally we state the equidistribution hypothesis implicit in the classical heuristics as a precise conjecture — that the block letters of uniformly sampled large starts carry, at every finite block length, the frequencies of an exactly computable Bernoulli law — prove its consequences conditionally, exhibit an unconditional base case inside the digit budget, and report a calibration campaign whose four apparent anomalies all dissolved under controls — one of them via an exact routing lemma that a biased estimator had been reflecting.

Version note

v1, July 2026 (original publication). v2, July 2026: restored the subtitle of the `merle` citation; added a note evidencing the sharpness hedge of Theorem 4.6 (Section 4), prompted by correspondence with Eric Merle. No theorem or universal claim is strengthened in v2, which adds

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a finite computational evidence record. v2-specific Zenodo DOI: 10.5281/zenodo.21421120. v3, August 2026 (drafted; the version-specific DOI below is reserved and this version is not yet published): a correction is appended to the Note added in v2 (Section 4) — the note's identification of the remaining gap has been superseded on both halves in the project record, and the correction states what closed, at what scope, where the record is, and that the proof is established there and not reproduced here. The original sentence stays in place and Theorem 4.6's statement is unchanged; its sharpness assessment, previously the theorem's closing clause, is set out beside it under *Sharpness evidence and assessment*, so that the theorem environment carries only what the paper proves. v3 also brings each definition and statement back into line with the project record. Definition 2.1 requires $\omega > 0$, without which $(-1, 1)$ leaves F undefined; Theorem 3.3's unconditional bound reads $C(\omega)(1+\log d)^2$, the printed $(\log d)^2$ being vacuous at $d = 1$; Theorem 3.8 states its depth-k window as the residues of Theorem 3.7 *together with* the stratum labels (s, σ, a_+), matching its own proof, with Theorem 3.7 unchanged; Theorem 4.5's $n_0(p)$ is defined; the digit budget is labelled a heuristic, as its own text always said, and no part of it is claimed as proved; and Proposition 4.1 defines σ_j at its point of use ($\sigma_j = s_j + m_{j+1}$, as in Definition 2.1) and distinguishes M_t from the anchor. Section 5 carries one object where it previously carried several. Its observable is the capped window $W_{k,D}$, at a depth k and a cap D quantified together, the cap bounding the stratum labels as well as the depth; the comparison law $\pi_{k,D}$ is the law of that window under the two-sided Bernoulli measure, its depth component the exact convolution rather than the window chain's stationary law, and "product" names its two proved clauses and no others — in particular the window process is stationary but not independent across time. Hypothesis 5.1 is stated in ensemble form — uniformly sampled starts, a horizon linked to the sampling scale, every block counted once, a single limit — in letter coordinates and at every finite block length, retiring both the single-orbit reading, which is empty on every convergent orbit, and the single-visit reading, which is strictly weaker; the distance is total variation on the finite window alphabet, named where it is used; and the bulk cut is replaced by an exponent budget whose admissibility is defined and whose protection is a deterministic identity rather than an assumption. Section 5 states no descent or contraction consequence of Hypothesis 5.1: that conclusion is a theorem without the hypothesis, and stronger, so the section is framed onto what the hypothesis alone supplies, which is distributional. Its unconditional base case is given with the exact rate at which it holds and the exact horizon at which that rate vanishes, and its density conclusion is stated at dyadic-shell scale, where it is exact — the union over all sampling scales being a triangular array that no per-scale statement controls. The stationary 3-adic law is attributed to Tao's Syracuse random variable and its negative-time reading to Tao's own footnote; the unconditional density line the hypothesis does not add to — Terras, Everett, Korec, Inselmann, Tao — is cited in Related work, together with the two further 3-adic studies (Wirsching, Thomas) that a reader of Tao's footnote will reach, with those horizons given in the step units in which they are proved, their reading in reduced blocks being a consequence of Hypothesis 5.1 rather than an input to it. The calibration record is reported with the two ceilings it does not reach past. Verification pointers and script names are concrete throughout. No numbered theorem's claim is strengthened, weakened, or renumbered, and nothing new is proved here; v3 reports two results established in the project record and not reproduced in this paper — the sharpness construction of Section 4 and the base-case assembly of Section 5. v3-specific Zenodo DOI (reserved): 10.5281/zenodo.21730505.

Author's note

Dear reader — I'm no mathematician, and I learned much of this paper's terminology during the process of formalizing it (p-adic valuations were completely new to me as a formal concept). It all began when I got hooked on a strange recursive pattern in the prime factors of a Collatz chain, and that idea pushed me into conversation with AI. We cleaned my idea up into the states (ω, d), with the primary goal of factoring as much deterministic behaviour as possible out of the question, in the hope of focusing attention where it matters. That simplification is what enables the paper's main contributions: exact laws for every single step; a uniform trim that caps what size-counting can achieve against cycles; and an exact hypothesis pointing the remaining question toward more fundamental "fair-coin" mathematics. Because this work is AI-assisted and because I am not formally trained in the relevant mathematics, Appendix A describes the verification protocol used throughout the project. The complete research record is public at https://github.com/macindoe/collatz. — Ben

1 Introduction

Let $T(x) = (3x+1)/2^{v_2(3x+1)}$ denote the odd-to-odd Collatz map on positive odd integers. The Collatz conjecture asserts that every T-orbit reaches 1. This paper develops a coordinate system in which the local arithmetic of T becomes exactly computable, and uses it for three purposes: to prove exact per-step laws for the induced dynamics; to determine how far counting arguments can go toward cycle exclusion (answer: exponentially far in the number of blocks, and provably no farther); and to state precisely the statistical hypothesis that the classical heuristics leave implicit.

The idea is to stop watching individual steps. Writing $x + 1 = 2^m u$ with u odd, the next $m - 1$ odd steps of T are deterministic — each converts one factor of 2 in $x + 1$ into a factor of 3 — and terminate in a single cascade (Proposition 2.2). Compressing this *block* and splitting $u = 3^a\omega$ with $3 \nmid \omega$ yields a state (ω, d) with $d = m + a$, and the block-to-block dynamics induce a self-map F on such states. The compression loses nothing the conjecture depends on (Theorem 2.3).

Contributions. (i) A reduced self-map F on states (ω, d), with an in-paper proof of well-definedness, whose convergence problem and cycle set are exactly equivalent to those of T (Section 2). (ii) A global anchor law for the exit valuation s, derived from one squaring lemma (Section 3). (iii) Exact residue-class formulas for the depth evolution, including an explicit 3-adic absorption law (Theorem 3.5). (iv) An exact low-order law for the anchor increment and an error-free trichotomy deciding the next step from a finite residue window together with the step's stratum labels, with a digit-budget accounting (Theorems 3.7–3.8, Heuristic 3.9). (v) A cycle elimination identity, short rederivations of the classical small-cycle exclusions, and — the main new theorems — a uniform trim with effective finiteness at every period together with a sharpness family showing the exponential loss is intrinsic to size-counting (Section 4). (vi) The equidistribution hypothesis stated exactly, its consequences proved conditionally, and a calibration record (Section 5).

Section 3 is the technical heart. The exit valuation, the 3-absorption, the next depth, and the low-order digits of the next anchor all obey exact laws in terms of the anchor displacement $d - M(\omega)$. A finite window of state digits and stratum labels then decides the next step in a trichotomy that never errs, and an elementary accounting argument shows each decision *consumes* digits irreversibly, indicating that bounded information cannot decide unbounded horizons. In these coordinates the difficulty of the problem is not diffuse: it is the question of where the digits of the anchors come from. Section 4 tests the one regime where finite data can beat unbounded depth — cycles — and proves both what counting can do there (effective finiteness, uniformly in the period) and that it can do no more (the staircase family). Since the staircase is precisely a divergent-orbit profile closed into a loop, the cycle half and the divergence half of the problem's residual difficulty are, in a concrete sense, the same object. Section 5 formalizes the statistical half: the folklore "fair-coin" heuristics (Terras [1]) become an exact statement about the block letters of typical orbits — equivalently about an explicitly computable window law $\pi_{k,D}$ — whose ledger consequences are unconditional theorems *about* $\pi_{k,D}$.

Related work and provenance

Block-like decompositions and stochastic models are classical (Terras [1]; see Lagarias [12] for the literature), and the 2-adic conjugacy of the Collatz map to the shift underlies all Haar-genericity heuristics. The unconditional density line descending from Terras is long and is not what Section 5 adds to. Terras [1] and Everett [2] place descent below the start at natural density one; Korec [3] lowers the target to y^c for every $c > \log_4 3 = 0.7924\ldots$, in natural density and within $O(\log y)$ steps; Inselmann [5] carries a two-sided trajectory law — the orbit tracked to within $m^{\pm\varepsilon}$ of its classical drift, simultaneously at every time up to $(\log_2 \frac{4}{3})^{-1} \log_2 m$ Syracuse steps, the full descent horizon in that unit — at natural density one, and so reaches m^ε for every fixed $\varepsilon > 0$; and Tao [4] reaches every divergent threshold, in logarithmic density. All of these horizons are counted in steps, and their reading in the reduced blocks of Section 5 is not free: it divides by a mean number of steps per block that is itself part of what Hypothesis 5.1 asserts. What Hypothesis 5.1 asserts is not descent but the letter statistics themselves — equivalently the exact window law $\pi_{k,D}$ — at horizons past the digit budget of Heuristic 3.9. Tao's paper carries a footnote to two further 3-adic studies, and neither is the object appearing here. Wirsching [6] completes the root of the predecessor tree 3-adically and studies the inverse map as a Markov chain on $[0, 1] \times \mathbb{Z}_3^\times$ in order to *count* predecessors; his inverse branches therefore carry equal weight and his limiting 3-adic law is Haar, whereas the law appearing in Section 5 weights each branch by the probability 2^{-a} that an orbit takes it and is consequently not uniform — $(\frac{2}{3}, \frac{1}{3})$ at level one where Haar gives $(\frac{1}{2}, \frac{1}{2})$. The two programmes are nonetheless siblings in shape: what Wirsching needs is that a Collatz-generated object be unbiased in 3-adic coordinates, and Hypothesis 5.1 is that one is unbiased in 2-adic coordinates. Thomas [7] proves that the integers whose orbit reaches 1 with all six frequencies over the units modulo 18 below $\frac{1}{6} + 0.0215$ have counting function at most $N^{0.9999}$; he constructs no limiting measure, and the law recorded here implies his conclusion, its level-one form already forcing some class to frequency at least $\frac{2}{9}$. Two-adic and Diophantine methods dominate the cycle literature [8, 10, 11, 9]; effective linear forms in logarithms [13, 14, 15] are the engine, and the verified range of the conjecture stands at 2^{71} [16]. Two contemporary human–AI efforts are close in spirit: a documented exploration [18] develops a burst–gap decomposition and an orbit equidistribution conjecture, and a Lean 4 formalization [17] proves cycle exclusion conditionally on Baker-type and verification hypotheses, including an impossibility lemma for measure-based uniform bounds kindred to our Theorem 4.6; neither contains the anchor machinery, exact per-step laws, or the counting-limit dichotomy developed here. A bachelor's thesis of Rackl (Klagenfurt, 2021) applies valuation methods to Syracuse cycles via the classical cycle equation; it likewise contains none of the present apparatus. The division of labor in the present work is stated precisely, because it matters for both credit and blame. The reduced coordinate system itself — the odd core ("seed"), the block cascade, the states (ω, d), and the driving question of Section 3 — is the human author's invention, originating in a hand-drawn note of 13 March 2023 that predates the collaboration. The analytical superstructure erected on it — the anchor formulation, the cycle apparatus, the equidistribution formalization, and the proofs — was developed in AI-assisted research under the author's direction, drawing on the classical literature's concepts; any overlap with contemporary work lies in that layer, not in the coordinate system. Appendix A states the responsibility and verification protocol.

2 The reduced system

Definition 2.1. *A* reduced state *is a pair (ω, d) with ω odd and positive, $3 \nmid \omega$, $d \geq 1$. Set*

$$A = 3^d\omega - 1, \qquad s = v_2(A)\,, \qquad x_{\text{exit}} = A/2^s, \qquad C = A + 2^s,$$

$$\sigma = v_2(C)\,, \qquad a_+ = v_3(C)\,, \qquad m_+ = \sigma - s, \qquad \omega_+ = \frac{C}{2^\sigma 3^{a_+}}, \qquad d_+ = m_+ + a_+,$$

and define the reduced map $F(\omega, d) = (\omega_+, d_+)$*. The* projection R *sends an odd x with $x + 1 = 2^m 3^a \omega$ ($3 \nmid \omega$) to $R(x) = (\omega, m + a)$.*

Proposition 2.2 (block structure and well-definedness)**.** *Let x be odd with $x+1 = 2^m u$, $u = 3^a\omega$ odd, $3 \nmid \omega$, and let $(\omega, d) = R(x)$, $d = m + a$. Then:*

(i) the next $m - 1$ iterates of T are x_j with $x_j + 1 = 2^{m-j}\,3^j u$ for $j = 0, \ldots, m - 1$;

(ii) the following iterate is $T(x_{m-1}) = A/2^s = x_{\text{exit}}(\omega, d)$, which depends only on (ω, d);

(iii) consequently $R(T^m(x)) = F(R(x))$ for every representative x of (ω, d), the representatives being exactly $x = 2^{d-a}3^a\omega - 1$, $0 \leq a \leq d - 1$; and the sequence of reduced states visited by any T-orbit is the F-orbit of its projection.

Proof. (i) If $y+1 = 2^k v$ with $k \geq 2$, v odd, then $3y+1 = 3(y+1)-2 = 2\left(3 \cdot 2^{k-1}v - 1\right)$, an odd multiple of 2, so $T(y) = 3 \cdot 2^{k-1}v - 1$ and $T(y) + 1 = 2^{k-1}(3v)$: one factor of 2 is exchanged for one factor of 3. Induction from $y = x$ gives (i). (ii) At $j = m - 1$ we have $x_{m-1} + 1 = 2 \cdot 3^{m-1}u$, so $3x_{m-1} + 1 = 3(x_{m-1} + 1) - 2 = 2(3^m u - 1) = 2A$ since $3^m u = 3^{m+a}\omega = 3^d\omega$. Thus $T(x_{m-1}) = 2A/2^{1+s} = A/2^s$, and A depends only on (ω, d). (iii) $x_{\text{exit}} + 1 = (A + 2^s)/2^s = C/2^s$, so $R(x_{\text{exit}}) = (\omega_+, (\sigma - s) + a_+) = F(\omega, d)$ by Definition 2.1; the representative list is immediate from $x + 1 = 2^{d-a}3^a\omega$. ☐

Theorem 2.3 (convergence translation)**.** *$(1, 1)$ is a fixed point of F whose R-fiber is $\{1\}$. The Collatz conjecture holds if and only if every F-orbit from a valid state reaches $(1, 1)$, and R induces a bijection between nontrivial T-cycles and F-cycles other than $\{(1, 1)\}$.*

Proof. For $(1, 1)$: $A = 2$, $s = 1$, $x_{\text{exit}} = 1$, $C = 4$, so $F(1, 1) = (1, 1)$; the fiber is $\{1\}$ since the representatives are $x = 2 \cdot 1 - 1 = 1$ only. If the conjecture holds and (ω, d) is valid, its representative $x = 2^d\omega - 1$ reaches 1, so by Proposition 2.2(iii) the F-orbit of (ω, d) reaches $R(1) = (1, 1)$. Conversely if every F-orbit reaches $(1, 1)$, then any odd x has a T-orbit visiting the fiber of $(1, 1)$, i.e. the value 1. For cycles: a nontrivial T-cycle projects to a periodic F-orbit avoiding $(1, 1)$ (its values avoid 1); conversely an F-cycle's block entries $e_t = x_{\text{exit}}(S_{t-1})$ are deterministic in the states and joined by genuine T-iteration, so e_0 is T-periodic and nontrivial; the two constructions are mutually inverse. ☐

3 Anchor dynamics: the exact per-step laws

For $u \equiv 1 \pmod 8$ the equation $9^t = u^{-1}$ has a unique solution $t \in \mathbb{Z}_2$, since 9 topologically generates $1 + 8\mathbb{Z}_2$; write $N(u)$ for it, so $N(u) = -\log u / \log 9$ in the 2-adic logarithm. N is an isomorphism $(1 + 8\mathbb{Z}_2, \times) \to (\mathbb{Z}_2, +)$, and

$$v_2\big(9^t - 1\big) = 3 + v_2(t) \qquad (t \in \mathbb{Z}_2 \setminus \{0\}), \tag{1}$$

the *isometry*, since $9 = 1 + 8$ and $v_2(\log(1 + 8z)) = 3 + v_2(z)$-type estimates hold with equality here.

Definition 3.1 (anchor)**.** *For every odd ω set $M(\omega) = N(\omega^2) = -2\log\omega/\log 9 \in \mathbb{Z}_2$, well defined since $\omega^2 \equiv 1 \pmod 8$. $M(\omega)$ is even iff $\omega \equiv \pm 1 \pmod 8$.*

Lemma 3.2 (squaring lemma)**.** *Let ω be odd, $d \geq 1$, and $Y = 3^d\omega$. If $Y \equiv 1 \pmod 8$, then*

$$v_2(Y - 1) = 2 + v_2(d - M(\omega))\,.$$

Proof. $Y^2 = 9^d\omega^2 = 9^d\, 9^{-M(\omega)} \cdot \big(9^{M(\omega)}\omega^2\big) = 9^{\,d - M(\omega)}$, since $9^{M(\omega)} = \omega^{-2}$ by definition of N. By (1), $v_2\big(Y^2 - 1\big) = 3 + v_2(d - M(\omega))$ (nonzero since $Y^2 \neq 1$ for a valid state). Since $Y \equiv 1 \pmod 8$ we have $v_2(Y + 1) = 1$, and $v_2(Y - 1) = v_2\big(Y^2 - 1\big) - v_2(Y + 1)$ gives the claim. □

Theorem 3.3 (global valuation law)**.** *Call (ω, d)* lifting *if $3^d\omega \equiv 1 \pmod 8$, i.e. $\omega \equiv 1\ (8)$ with d even or $\omega \equiv 3\ (8)$ with d odd. On the lifting classes*

$$s = v_2\Big(3^d\omega - 1\Big) = 2 + v_2(d - M(\omega)) \;\geq\; 3,$$

while on the six non-lifting residue-parity classes $s \in \{1, 2\}$ is the constant determined by ($\omega \bmod 8$, $d \bmod 2$): $s = 1$ on $(1, \mathrm{odd}), (5, \mathrm{odd}), (7, \mathrm{even}), (3, \mathrm{even})$ and $s = 2$ on $(5, \mathrm{even}), (7, \mathrm{odd})$. Unconditionally $s \leq C(\omega)\,(1 + \log d)^2$ by p-adic Baker theory [13, 14].

Proof. The lifting case is Lemma 3.2; the parity claim ($d \equiv M(\omega) \bmod 2$ on lifting classes) follows from $v_2(d - M(\omega)) \geq 1$ there. Off-lifting, $3^d\omega \bmod 8 \in \{3, 5, 7\}$ gives $v_2\big(3^d\omega - 1\big) \in \{1, 2\}$ by inspection of residues mod 8. The Baker bound applies to $s = v_2(9^n\tilde{\omega} - 1)$, a linear form in two 2-adic logarithms. □

Lemma 3.4 (3-adic absorption law)**.** *Let $h(s) = v_3(2^s - 1)$, so $h(s) = 0$ for s odd and $h(s) = 1 + v_3(s)$ for s even. Then*

$$a_+ = v_3(C) = \begin{cases} 0, & s \text{ odd}, \\ \min(d,\, h(s)), & s \text{ even},\ d \neq h(s), \\ d + v_3\big(\omega + (2^s - 1)3^{-d}\big)\,, & s \text{ even},\ d = h(s). \end{cases}$$

In particular the next state gains a factor of 3 exactly when s is even.

Proof. $C = 3^d\omega + (2^s - 1)$ with $v_3\big(3^d\omega\big) = d$ exactly and $v_3(2^s - 1) = h(s)$; $h(s) = 0$ iff s odd since $2^s \equiv 2 \pmod 3$ then, and for even s, lifting-the-exponent gives $v_3\big(4^{s/2} - 1\big) = v_3(3) + v_3(s/2) = 1 + v_3(s)$. The ultrametric equality $v_3(X + Y) = \min(v_3(X)\,, v_3(Y))$ holds when the valuations differ; when $d = h(s)$, factor 3^d and evaluate the remaining unit sum. □

Theorem 3.5 (depth closure)**.** *On every residue class the next depth is $d_+ = m_+ + a_+$ with a_+ given exactly by Lemma 3.4 and m_+ by the* entry-depth law*: writing ε for the anchor displacement ($\varepsilon = n - N(\omega)$ on the lifting class $d = 2n$, $\varepsilon = n - N(3\omega)$ on $d = 2n+1$) and $\delta_s = N(1-2^s)$ (which satisfies $v_2(\delta_s) = s-3$ for $s \geq 3$),*

$$m_+ = 3 - s + v_2(\varepsilon + \delta_s) \quad \textit{(lifting)}, \qquad m_+ = \begin{cases} 1 + v_2(d - M(\omega)), & s = 1, \\ v_2((d-1) - M(\omega)), & s = 2. \end{cases}$$

The forced carry $v_2(\delta_s) = s - 3 = v_2(\varepsilon)$ yields $m_+ \geq 1$ structurally.

Proof. All cases reduce to the *target-shift lemma*: for $c \in 1 + 8\mathbb{Z}_2$ and $d = 2n$ with $\omega \equiv 1$ (8),

$$v_2\Big(3^d\omega - c\Big) \;=\; 3 + v_2(n - N(\omega) + N(c)),$$

proved exactly as Lemma 3.2 after writing $3^d\omega - c = 9^{-N(c)}\big(9^{\,n-N(\omega)+N(c)} - 1\big)$. Since $m_+ = v_2(x_{\text{exit}} + 1)$ and $2^s(x_{\text{exit}}+1) = C = 3^d\omega - (1-2^s)$, the lifting case is the shift $c = 1 - 2^s$ (which lies in $1 + 8\mathbb{Z}_2$ for $s \geq 3$), with $v_2(\delta_s) = s - 3$ from the isometry; the odd lifting component reduces to the even one via the companion parameter 3ω. For $s = 1$: the shifted target is -1, and $v_2(X+1) = v_2\big(X^2-1\big) - v_2(X-1)$ with $X = 3^d\omega$, $X^2 = 9^{\,d-M(\omega)}$, gives the formula; For $s = 2$: here $C = 3^d\omega + 3 = 3(X'+1)$ with $X' = 3^{d-1}\omega$, so $m_+ = v_2(C) - 2 = v_2(X'+1) - 2$. The $s = 2$ classes force $3^d\omega \equiv 5 \pmod 8$, hence $X' \equiv 3^{-1}\cdot 5 \equiv 7 \pmod 8$, so $v_2(X'-1) = 1$; and $X'^2 = 9^{\,(d-1)-M(\omega)}$ as in Lemma 3.2, so $v_2(X'+1) = v_2\big(X'^2-1\big) - v_2(X'-1) = \big(3 + v_2((d-1) - M(\omega))\big) - 1$, giving $m_+ = v_2((d-1) - M(\omega))$ as claimed. ☐

Remark 3.6 (verification notes). Each law above was checked by independently written code, with zero discrepancies. The valuation law on 8,000 random states across both lifting components, and the entry-depth law exhaustively on all 62,937 valid states with $\omega < 3000$, $d < 64$; both are recorded at `stage3.md` §11.8.6.3 of the project record and in that page's status header. The absorption law on 327,980 states — all 266,680 valid states with $\omega < 20{,}000$, $d \leq 40$, plus 60,000 random states with $\omega < 2^{64}$, $d \leq 200$, plus 1,300 states built to order — checked against both this Lemma and its form in the record, 983,954 comparisons in all. Because a_+ is a 3-adic function of ω on the boundary branch $d = h(s)$, that branch is constructed rather than sampled: 3,196 of the states lie on it, 880 of them built explicitly across $d \in \{1, 2, 3\}$, and the shallow branch $d < h(s)$ likewise (533 states, 420 built). Code: `experiments/absorption_law.py`.

Theorem 3.7 (low-order anchor-increment law)**.** *Let $\Delta M = M(\omega_+) - M(\omega)$. For every $k \geq 1$, $\Delta M \bmod 2^k$ is determined by, and explicitly computable from,*

$$\omega \bmod 2^{\sigma+k+2}, \qquad d \bmod 2^{\sigma+k}, \qquad a_+ \bmod 2^k,$$

and the stratum data (s, σ) are determined by the same residues. This last clause does not extend to a_+: by Lemma 3.4 the absorption is a 3-adic function of ω, and it enters the display only through its residue $a_+ \bmod 2^k$.

Proof. Three digit-determinacy facts chain together. (a) For $u \equiv 1$ (8), $N(u) \bmod 2^k$ depends only on $u \bmod 2^{k+3}$, since 9 has order 2^k mod 2^{k+3} and $9^t \equiv u^{-1}$ determines $t \bmod 2^k$. (b) $C \bmod 2^q$ is determined by $\omega \bmod 2^q$, $d \bmod 2^{q-2}$, s, since 3 has order 2^{q-2} mod 2^q. (c) $\omega_+ \bmod 2^r$ is determined by $C \bmod 2^{\sigma+r}$, σ, and $a_+ \bmod 2^{r-2}$, by exact division and the order of 3 mod 2^r. Now $\Delta M = N\big((\omega_+/\omega)^2\big)$; by (a) this needs $(\omega_+/\omega)^2 \bmod 2^{k+3}$, hence $\omega_+, \omega \bmod 2^{k+2}$; by (c) with $r = k+2$ this needs $C \bmod 2^{\sigma+k+2}$ and $a_+ \bmod 2^k$; by (b) this needs the stated residues of ω and d. Finally $s \leq \sigma$ and σ are visible in $A \bmod 2^{\sigma+1}$, $C \bmod 2^{\sigma+1}$. ☐

Theorem 3.8 (one-step trichotomy)**.** *The* depth-k window *of a valid state* (ω, d) *consists of the residues of Theorem 3.7 together with the stratum labels* (s, σ, a_+)*. From the depth-k window alone, one can determine: the exact next depth* d_+ *and next residue-parity class; the exact value* $s_+ \in \{1, 2\}$ *if the next class is non-lifting; and, if lifting, the truncation of the next displacement mod* 2^k *— yielding* s_+ *exactly when nonzero, and the correct report "*$s_+ \geq k + 2$*" when zero. The window never outputs a wrong value; under the law* $\pi_{k,D}$ *of Section 5 the undecided rate is* $\approx 2^{-(k+1)}$.

Proof. By facts (b), (c) above the window determines $\omega_+ \bmod 2^{k+2}$, hence the next class; $d_+ = (\sigma - s) + a_+$ is exact from the stratum data. Non-lifting s_+ is the class constant (Theorem 3.3). Lifting: $M(\omega_+) \bmod 2^k$ follows from fact (a) applied to ω_+^2, so $(d_+ - M(\omega_+)) \bmod 2^k$ is computable; if nonzero its valuation is visible and Theorem 3.3 gives s_+; if zero then $v_2(d_+ - M(\omega_+)) \geq k$, i.e. $s_+ \geq k + 2$. □

Heuristic 3.9 (digit budget)**.** *Deciding one step at depth* k *consumes the state's 2-adic data to depth* $\sigma + k + 2$ *(empirical mean* $\sigma \approx 4.0$ *along orbits), and nothing regenerates it: the anchors of later states are functions of the full initial state, not of any bounded window. A window of* W *digits therefore supports* $O(W/(k + 6))$ *decided steps. This strongly suggests — and we treat it as the organizing heuristic, not a formalized theorem — that bounded-window determinism cannot decide unbounded horizons without additional statistical or rigidity input.*

Remark 3.10 (verification note). The trichotomy was run along real orbits for $k \in \{4, 8\}$: 21,296 steps, 21,000 decided with zero errors, 296 flagged deep with zero violations of the bound $s_+ \geq k + 2$. Code: `experiments/one_step_propagation.py`; the low-order law of Theorem 3.7 is checked by `experiments/anchor_increment.py`.

Heuristic 3.9 is the paper's organizing negative observation: it splits the residual difficulty into a statistical question for typical starting values (Section 5) and a rigidity question for cycles (Section 4); we know of no residual content outside those two.

4 Cycles: rederivations, a uniform trim, and its sharpness

Fix a hypothetical p-block cycle of F with entry depths $m_t \geq 1$, exit valuations $s_t \geq 1$ ($t \in \mathbb{Z}/p\mathbb{Z}$), and set $n = \sum m_t$, $K = \sum s_t + n$, $q = 2^K - 3^n$.

Proposition 4.1 (elimination identity)**.** $q > 0$*, and for every rotation* r*,*

$$\omega_r \, 3^{a_{r-1}} \, q \;=\; R_r \;:=\; \sum_{t=0}^{p-1} 3^{M_t} \, 2^{S_t} \, (2^{s_t} - 1),$$

where, in rotation order starting at r*,* $M_t = \sum_{j>t} m_j$ *and* $S_t = \sum_{j<t} \sigma_j$*, with* $\sigma_j = s_j + m_{j+1}$*, indices cyclic (the step's* $\sigma = v_2(C)$*,* $m_+ = \sigma - s$*, of Definition 2.1; the shift is essential,* m_{j+1} *and not* m_j*; and* M_t*, a partial sum of entry depths, is unrelated to the anchor* $M(\omega)$ *of Section 3 — this is the project record's own notation,* `cycles.md` *§12.6.1, kept so that paper and record read together). Hence* $q \mid R_r$ *and* $q \leq \min_r R_r$*. For the trivial cycle,* $R = 4^p - 3^p = q$ *identically.*

Proof. The step identity $2^{\sigma_t} 3^{a_t} \omega_{t+1} = 3^{d_t} \omega_t + (2^{s_t} - 1)$ (Definition 2.1) unrolls p times; imposing closure $\omega_p = \omega_0$, substituting $d_j = m_j + a_{j-1}$, and cancelling the common power of 3 yields the display. Positivity: the right side is positive. □

Corollary 4.2 (size balance). *$2^K = 3^n \prod_t (1+\varepsilon_t)$ with $\varepsilon_t = (2^{s_t}-1)/(3^{d_t}\omega_t)$; if every cycle element exceeds X then $0 < K\log 2 - n\log 3 < p/X$. This is the classical Baker-side engine [9], obtained in four lines.*

Lemma 4.3 (ceiling). *If $K > \lceil n\log_2 3\rceil$ then $\prod(1+\varepsilon_t) \ge 2$, so some $\varepsilon_t \ge 2^{1/p}-1 \ge (\ln 2)/p$ and some block exit is $< p/\ln 2$. Consequently $K = \lceil n\log_2 3\rceil$ for every nontrivial cycle with $p \le \ln 2\cdot X_0$, X_0 the verified range [16] — unconditionally for $p \le 5$ by inspecting the odd values ≤ 7.*

Theorem 4.4 (periods 1–3; rederivations). *F has no nontrivial cycles of 1, 2, or 3 blocks. These statements are contained in the classical results [8, 10, 11]; the derivations, not the theorems, are the contribution.*

Proof outline. $p=1$: the fixed-point equation is $\omega\, 3^a(2^{s+m}-3^m) = 2^s-1$; the sign and size constraints pin 2^s into a window of endpoint ratio $(1-3^{-m})/(1-2^{-m}) < 2$, leaving at most one candidate s per m; exact search kills $m \le 20{,}000$ and Steiner's theorem [8] covers all m via the block–circuit correspondence. $p = 2,3$: budget-trim lemmas (casework over rotations' dominant terms, using $\sum s_t = K-n$) force $0 < K\log 2 - n\log 3 < 2^{6-n/5}$ resp. $2^{3-0.115n}$; exact searches to $n = 20{,}000$ find no solutions; any effective irrationality measure for $\log_2 3$ [15] excludes the tail. Full details are in the project record: the three classifications at `cycles.md` §12.2.3, §12.5.3 and §12.7.5, the two trim lemmas at §12.5.2 and §12.7.4, and the ceiling lemma that supplies $K = \lceil n\log_2 3\rceil$ at §12.6.2. The searches are `experiments/period1_cycles.py`, `experiments/period2_cycles.py` and `experiments/period3_cycles.py`. □

Theorem 4.5 (uniform trim). *Every nontrivial p-block cycle satisfies, with $\gamma := K - \log_2 q$,*

$$\gamma + \log_2 p \;>\; \frac{(\log_2 3 - 1)\, n}{(\log_2 3)^{\,p} - 1}.$$

Consequently the candidate set at every period is finite and explicitly bounded. Combined with Rhin's effective bound [15],

$$K\log 2 - n\log 3 \;>\; \exp\bigl(-13.3\,(0.46057 + \log n)\bigr),$$

every nontrivial p-block cycle has $n \le n_0(p)$, the unique solution in n of

$$\frac{(\log_2 3 - 1)\, n}{(\log_2 3)^{\,p} - 1} \;=\; \log_2 p \;+\; \frac{p + 13.3\,(0.46057 + \log n)}{\log 2},$$

so that $n_0(p) = O(p\,(\log_2 3)^p)$.

Proof. For each rotation, $q \le R_r < p\, 2^{\max_t e_t}$, where the exact exponent identity gives $e_t - K = (\log_2 3 - 1)M_t - m_r - \sum_{j>t} s_j$. The resulting conditions $m_r < \gamma + \log_2 p + \Phi_r$ close into the cyclic recursion $\Phi_r = \max(0, \Phi_{r-1} + w_{r-1})$ on arc weights $w_j = (\log_2 3 - 1)m_j - s_j$, whose cyclic total $(\log_2 3 - 1)n - \sum s_t$ is negative because $2^K > 3^n$ forces $\sum s_t > (\log_2 3 - 1)n$. A max-plus recursion with negative cyclic total must vanish somewhere (otherwise following it once around the cycle would add the total to itself); fix such a reset index r_0, so $\Phi_{r_0} = 0$. Walking forward, $\Phi_{r_0+j} \le (\log_2 3 - 1)T_{j-1}$ where T_j is the partial sum $m_{r_0} + \cdots + m_{r_0+j}$ (drop the $-s$ terms), so the block bound gives $m_{r_0+j} < (\gamma + \log_2 p) + (\log_2 3 - 1)T_{j-1}$, i.e. $T_j < \log_2 3\, T_{j-1} + (\gamma + \log_2 p)$. Iterating p times from $T_{-1} = 0$ yields $n = T_{p-1} < (\gamma + \log_2 p)\,\bigl(\log_2 3^{\,p} - 1\bigr)/(\log_2 3 - 1)$, which is the display. □

Theorem 4.6 (sharpness: the staircase)**.** *No trim uniform in p can extend the small-period constants: there exist configurations satisfying every rotation's exact size condition $q \le R_r$ whose γ falls far below any polynomial-in-p extension of the constants of periods 2–3. Explicitly, at $p = 7$, $n = 94$, the staircase $m = (4, 7, 9, 15, 23, 35, 1)$ with $s = (1, \ldots, 1,\ S - 6)$ passes all seven size tests with $\gamma = 6.74$, where the period-3 constant would demand $\gamma > 8.9$; 84 further verified instances occur at $p = 6$ alone. All such configurations fail the divisibility conditions $q \mid R_r$. Uniform cycle exclusion therefore requires the divisibility system — equivalently, rigidity of the closed anchor walk $\sum_t \Delta M_t = 0$.*

Sharpness evidence and assessment. The construction behind these witnesses — block depths growing geometrically at ratio $\approx \log_2 3$ with unit exit valuations, closed by a single block of unit depth and maximal exit valuation — tracks the extremal configuration of the max-plus recursion in the proof of Theorem 4.5. Supported by the verified instances, though not proved here for all p, we assess that it passes all size conditions with $\gamma = O(\log p)$ for every p. This assessment is evidence for Theorem 4.6 and no part of it: the theorem is closed at $p = 6, 7$ by the witnesses exhibited above. The assessment has since been proved in the project record, at a scope and a shape stronger than assessed here; the Note added in v2 and its correction below state what was proved, where, and at what scope.

Remark 4.7. The staircase profile — geometric depth growth with shallow exits — is exactly the growth regime of divergent orbits. Size analysis cannot forbid it as a cycle for the same reason drift analysis cannot forbid divergence. The two halves of the residual difficulty meet in one configuration.

Note added in v2 (July 2026)

Prompted by correspondence with Eric Merle concerning this theorem's sharpness hedge (his related formal work is cited in [17]), we attempted a single period-parametrized construction procedure toward the assessed claim of Theorem 4.6: semiconvergents of $\log_2 3$ select the exponent n, a rounded geometric profile builds the climb, and a bounded correction closes the last bits, applied separately at each period. Exact big-integer verification by this recipe produced a passing size-condition witness ($q \le R_r$ at every rotation) at every period

$$p \in \{2, \ldots, 23\}, \qquad \gamma / \log_2 p \in [1.828,\ 3.643]$$

— now a contiguous range, extending the two isolated instances above. Initially the recipe's own candidate chain left $p = 22$ unresolved; correspondence with Eric Merle identified the cause as a gap in that chain's coverage at the required scale, not a failure of the correction step, and at candidates outside the chain ($n = 25217$, $n = 31202$) the same profile-and-correction procedure resolves it (13 and 8 correction moves), closing the range above. The remaining gap is the one already named: no proved closed-form bound on the multiplicative gap between consecutive correctly-signed semiconvergent runs — the bound that would certify no period is skipped — and the $p = 22$ episode is a demonstration that this gap bites in practice, not only in principle. The hedge sentence above is therefore unchanged: finite-range evidence supports the assessed $\gamma = O(\log p)$ behaviour, but does not prove it for all p. The construction, the verified instance record, and the diagnosis of the remaining gap are recorded at `cycles.md` §12.8.6 of the project's public repository (`https://github.com/macindoe/collatz/blob/72ec88e/cycles.md`).

(Correction, 2026-08-01.) The gap named above has since been closed, and closed by replacing the route rather than completing it. Candidate availability needs no continued-fraction input: because $8 - 5\log_2 3 = 0.0751874964\ldots$ is positive and below the target arc's length, any 66 consecutive integers contain an n whose $\lceil n\log_2 3\rceil - n\log_2 3$ lies in $[0.0415,\ 0.1169390665\ldots]$, and the scale window at period p holds $0.05\,(\log_2 3)^p$ integers — 79 at $p = 16$, growing geometrically — so $p = 16$ is the first period supplying 66 consecutive integers. The bound on the multiplicative gap between correctly-signed runs is not needed, and as posed it is a dead end: those gaps are the partial quotients of $\log_2 3$, so a uniform bound on them is exactly the assertion that $\log_2 3$ is badly approximable. The $p = 22$ episode was a property of the candidate list used, not of $\log_2 3$: under the availability statement the episode does not arise — from $p = 8$ upward not one working witness is a convergent or semiconvergent denominator — and the corrected construction supplies a passing size-condition witness at every period tested (end to end at $p \in \{3, \ldots, 26\}$; the construction itself verified through $p = 32$). The construction half is closed independently: a corrected profile — the geometric climb with a fixed additive offset $1/(\log_2 3 - 1) = 1.70951$ per block, absent from the profile described above — satisfies all p size conditions by construction, with no correction step, so the bounded correction is removed from the argument rather than bounded. The two halves compose to a proof for every period $p \geq 16$, at γ between the absolute constants 3.683012 and 5.140212 — no p-dependence at all — with $3 \leq p \leq 15$ meeting the same bracket by finite check, and $p \in \{2, 4\}$, outside the construction's reach, covered by direct exhibition. That proof is established in the project record at the reference below and is *not reproduced in this paper*; Theorem 4.6 stands above exactly as written, and what is proved in the record is the assessment stated above under *Sharpness evidence and assessment*, at a stronger shape than that assessment claims. Every constructed configuration remains a size-passer only and fails the divisibility conditions $q \mid R_r$, as the instances above do: sharper evidence that counting cannot do better, and no evidence about exclusion. The current record is `cycles.md` §12.8.6 of the project's public repository (`https://github.com/macindoe/collatz/blob/9d9d1ec/cycles.md`).

5 The equidistribution hypothesis, made exact

What is the right comparison object? By Theorem 3.8, all dynamical decisions at horizon one are functions of the depth-k window; the natural null model is therefore a measure on windows, and the natural choice of measure is forced by the structure: the ω-residues carry the anchor digits, for which Haar measure is the canonical reference (this is where the classical 2-adic shift-conjugacy heuristics live), while the depth d is *dynamical* — it is produced by Theorem 3.5 — and so receives not Haar measure but the exact renewal law of the depth itself: $d_+ = m_+ + a_+$ with m_+ geometric(1/2) and *independent* of the absorption a_+, so that it is the distribution of the depth, and not that of the absorption, which is the convolution of the two. The stationary 3-adic law governing a_+ (`aeh.md` §13.6.5) is not new here: it is Tao's Syracuse random variable $\mathrm{Syrac}(\mathbb{Z}_3)$ [4, Lemma 1.12 and Remark 1.13], in the present normalisation. Precisely, the 3-adic past-limit of the block coordinate has the law of $\mathrm{Syrac}(\mathbb{Z}_3)/2$ — the two differ by the 3-adic unit 2^{-1} because one block here is a run of Syracuse steps terminated by the first exponent ≥ 2, so that the block chain sees the Syracuse chain conditioned on that event — and hence $a_+ = v_3(\mathrm{Syrac}(\mathbb{Z}_3) + 2)$. The values $P(a_+ = 0) = \frac{2}{3}$, $P(a_+ = 1) = \frac{19}{63}$, $P(a_+ \geq 2) = \frac{2}{63}$, and thence $P(d = 1) = \frac{1}{3}$, $P(d = 2) = \frac{20}{63}$, are read off the nine values of $\mathrm{Syrac}(\mathbb{Z}/9\mathbb{Z})$ that Tao computes. What the present coordinates add is the *joint labelled* law (`aeh.md` §13.6.3(v)): the ω-

residues are Haar-uniform among odd residues and independent of the depth, with $d_+ = m_+ + a_+$ and $m_+ \perp a_+$, the whole carried on the residues *together with* the stratum labels (s, σ, a_+) — a_+ being a 3-adic function of ω (Lemma 3.4) which the finite 2-adic residue window does not determine. Two integers fix the observable: a depth k and a *cap* D, chosen together and quantified over. The *capped depth-k window* at a visit is

$$W_{k,D} \;=\; \big(\omega \bmod 2^{k+2},\; d \wedge D,\; s \wedge D,\; \sigma \wedge D,\; a_+ \wedge D\big), \qquad u \wedge D := \min(u, D),$$

a finite alphabet of at most $2^{k+1}D^3(D+1)$ letters; the cap is what bounds it, and it must cap the labels as well as the depth, since σ and a_+ are unbounded. This is *not* the window of Theorem 3.8: that object carries residues to the state-dependent depth $\sigma + k + 2$ together with the *exact* labels, so its alphabet is countably infinite, and it decides the next step. $W_{k,D}$ decides nothing; it is a statistic, and no statement in this section asks more of it. Let $\pi_{k,D}$ denote the law of $W_{k,D}$ under the two-sided Bernoulli measure $\hat{B} = \bigotimes_{i \in \mathbb{Z}}(\mathrm{geom}(\frac{1}{2}) \times \mathrm{geom}(\frac{1}{2}))$ on the door-letter alphabet — two-sided because the absorption a_+ is a function of the orbit's 3-adic past and has no realisation on the one-sided 2-adic side (`aeh.md` §13.6.3(iii), §13.6.5); reading the stationary Syracuse variable as the outcome of an iteration extending to arbitrarily large negative times is Tao's own alternative to his positive-time indexing [4, Remark 1.13, footnote 4 of arXiv v7]. "Product" names exactly two clauses of $\pi_{k,D}$ — residue $\perp$ depth, and $m_+ \perp a_+$ — and no others: $\pi_{k,D}$ is not a product across its coordinates (s is an exact function of the full state), and the process $(W_{k,D}(n))_n$ is stationary but not independent across time (e.g. from $(\omega \equiv 1\ (8),\ d$ odd) the next depth is exactly 1). The stationary law of the exact window chain is a $\sim 1\%$-accurate model of the depth marginal of $\pi_{k,D}$ rather than the marginal itself — the resolution at which `aeh.md` §13.4 recorded it — and the two differ by exactly computed rationals (`aeh.md` §13.6.5). The class skeleton of the chain is computable exactly: the transition from $(\omega \equiv 1\ (8),\ d$ odd) cited above holds because $m_+ = 1$ and $a_+ = 0$ there, and from $(\omega \equiv 5\ (8),\ d$ odd), $P(d_+ \text{ even}) = \frac{2}{3}$ exactly. Under $\pi_{k,D}$, unconditionally: $P(s = j) = 2^{-j}$ at every $j < D$ (four of eight classes give $s = 1$, two give $s = 2$, and the lifting shells are geometric), the mass beyond the cap sitting in one tail cell; the 3-gain rate is $\sum_{j \text{ even}} 2^{-j} = \frac{1}{3}$ by Lemma 3.4, exact up to that same cell, whose parity split the cap hides; and the classical negative drift is a property of $\pi_{k,D}$ — but only of $\pi_{k,D}$: window equidistribution at each fixed (k, D) does not control the means of the unbounded m_+ and s, so no drift or contraction statement about orbits follows from it (`aeh.md` §13.3.2). The folklore "ledger" is thus a theorem *about* $\pi_{k,D}$; the empirical question is only whether the orbits of typical starting values follow $\pi_{k,D}$.

Hypothesis 5.1 (AEH, ensemble form)**.** *Fix a* budget rate $\tau > 0$ *and a* block horizon rate $\theta > 0$*; for each N put $b = \lfloor \log_2 N \rfloor$, $\Lambda_N = \lceil \tau b \rceil$ and $T_N = \lceil \theta b \rceil$. Draw x uniformly from the odd integers of $[N, 2N)$; put $(\omega_0, d_0) = R(x)$ and $(\omega_{n+1}, d_{n+1}) = F(\omega_n, d_n)$; let the* letter *at block n be $\ell_n = (m_{+,n},\, s_{n+1})$, and let $S_n = \sum_{i<n}(m_i + s_i)$ be the* exponent spent *before block n — the number of 2's divided out from x to $x_{\mathrm{exit}}(n-1)$. Tally block n at the symbol*

$$\tilde{\ell}_n \;=\; \ell_n \;\; \text{if } S_n < \Lambda_N, \qquad \tilde{\ell}_n \;=\; \dagger \;\; \text{otherwise.}$$

For a finite word $w = (w_1, \dots, w_L)$ of letters, let $f_N(w, x)$ be the frequency of w among the $T_N - L + 1$ blocks $(\tilde{\ell}_n, \dots, \tilde{\ell}_{n+L-1})$, $0 \le n \le T_N - L$: every block counted exactly once, at weight $1/(T_N - L + 1)$ — the same number for every block of every orbit, so no block is reweighted by the orbit it came from and no denominator is random. Then for every finite word w and every $\varepsilon > 0$,

$$\frac{2}{N}\,\#\big\{\, x \text{ odd},\; N \le x < 2N \;:\; \big|f_N(w, x) - B[w]\big| > \varepsilon \,\big\} \;\longrightarrow\; 0 \qquad (N \to \infty),$$

where, writing $w_i = (m_i, r_i)$ for the components of the i-th letter of w, $B[w] = \prod_i 2^{-(m_i+r_i)}$, and $B[w] = 0$ for any w containing †, for every admissible pair (τ, θ) in the sense defined below.

Equivalently, and this is the form the calibration measures: for every k, D and L, the empirical distribution of the L-blocks of consecutive capped windows $W_{k,D}$ over the same tallied blocks, with † carried through and given mass 0 by $\pi^{(L)}_{k,D}$, converges in total variation on the finite window alphabet, off a vanishing density of starts, to $\pi^{(L)}_{k,D}$, the L-block law of the stationary process under $\hat{B}$. The equivalence is Theorem 13.6.4 of `aeh.md`, a deterministic dictionary between letters and labelled window blocks; the case $L = 1$ recovers $\pi_{k,D}$, and $\pi^{(L)}_{k,D} \neq \pi^{\otimes L}_{k,D}$. A single sampled orbit segment has no infinite past at its first block, so the reconstruction of $W_{k,D}$ from letters is exact only away from the segment's start; the deviation occupies $O(1)$ blocks of a horizon $T_N \to \infty$ and contributes $O(1/T_N)$ to every frequency above.

There is one limit here, $N \to \infty$, and the sample grows because the sampling scale grows rather than because any one orbit is run forever. That is forced. Along a fixed orbit no limiting procedure survives: the unrestricted empirical distribution is false on every convergent orbit, whose tail sits at $(1, 1)$ forever; and above a fixed cut a convergent orbit supplies only finitely many qualifying visits, and none at all once the cut exceeds its maximum, so a limit taking orbit length first and the cut second is empty rather than merely delicate, in whichever order it is taken.

The horizon does the job a bulk cut would do, and does it without one. A step of the one-division map $y \mapsto y/2$ or $(3y + 1)/2$ never lowers $\log_2$ by more than 1, so $\log_2 x_{\text{exit}}(n - 1) \geq \log_2 x - S_n$ for every start and every n, with no exceptional set; inside a budget $\tau < 1$ of the start's own bits every tallied exit therefore exceeds $N^{1-\tau}$, far above the *bottom regime* of `aeh.md` §13.1 — the fixed drainage basin of small integers, whose window statistics are the digits of particular numbers rather than samples from a measure. That matters because any altitude threshold is a selection on the observable itself: $x_{\text{exit}} = (3^d\omega - 1)/2^s$, so a cut on it censors large s, and a cut on the core ω_+ censors large s, m_+ and a_+ together. The budget clause $S_n < \Lambda_N$ is instead *predictable* — decided by blocks strictly earlier than the one it admits — and the tally denominator is the deterministic T_N that `aeh.md` §13.5's standing rule — fixed horizon, unweighted, per-visit sampling from uniform starts — was written to secure. Call τ *protected* when all but a vanishing density of starts keep every in-budget exit above any X_N with $\log X_N = o(\log N)$, and (τ, θ) *consistent* when $\theta\, \mathbb{E}_B[m + r] < \tau$; *admissible* means both. Every $\tau < 1$ is protected outright by the bound above, and every $\tau < (1 - \log_2 \sqrt{3})^{-1} = 4.8188\ldots$ is protected at natural density one by Inselmann [5, Thm. 1.10] — the same unit, nothing converted. Consistency is a theorem where the cylinder count runs and is part of Hypothesis 5.1 where it does not, since $\pi_{k,D}$ gives † no mass; it says exactly that blocks average $\mathbb{E}_B[m + r] = 4$ of exponent. Hence $4\theta < \tau < 1$ is $\theta < 1/4$ and $4\theta < \tau < 4.8188\ldots$ is $\theta < 1/\beta = 1.2047\ldots$, where $\beta = 2(2 - \log_2 3) = 0.8301\ldots$ is the classical per-block contraction rate and $4/\beta = (1 - \log_2 \sqrt{3})^{-1}$ is an identity: the block reading of either threshold divides by a mean the hypothesis itself supplies.

The hypothesis has an unconditional base case, and it is Heuristic 3.9 with a number on it. The classical coding fact — the odd integers whose first n blocks realize a prescribed itinerary form exactly one residue class modulo 2^{S+1}, S the itinerary's total exponent (Terras [1]; in the present coordinates `itinerary.md` §14.15.1.5) — makes the first n blocks of a start drawn uniformly from *any* window of 2^J consecutive integers exactly product-distributed on the event $S + 1 \leq J$: the event that the itinerary has not outspent the start's supply of binary digits.

Writing $b = \lfloor \log_2 N \rfloor$, this gives, for every $J \geq 2$,

$$\left\| \mathrm{Law}(\ell_0, \ldots, \ell_{n-1}) - B^{\otimes n} \right\|_{\mathrm{TV}} \;\leq\; \frac{2^{J+2}}{N} + P_B(S_n \geq J), \quad P_B(S_n \geq J) = P\big(\mathrm{Bin}(J-1, \tfrac{1}{2}) < 2n\big),$$

the identity on the right because S_n is the waiting time for the $2n$-th head in a fair coin sequence. The first term is the price of a general window $[N, 2N)$ over a dyadic one and is negligible for $J \leq (1-\eta)b$; the second decays at the exact rate $\log 2 - H(2\theta)$ per bit at $n = \lceil \theta b \rceil$, H the binary entropy — positive for every $\theta < 1/4$ and *zero* at $\theta = 1/4$, since the optimal Chernoff tilt is $e^\lambda = 2(1-2\theta)$ and needs $\lambda > 0$. The bound is on the joint law of the whole length-n word, so it controls every finite block length at once. Together with the concentration of empirical pattern frequencies, the removal of the segment's initial past-boundary — $O(1)$ blocks of a horizon $T_N \to \infty$ — and the altitude bound above, which makes the budget its own protection, this yields the hypothesis outright: Hypothesis 5.1 is a *theorem* for every admissible (τ, θ) with $\tau < 1$, equivalently for every horizon rate $\theta < 1/4$, at every block length. The assembly is Lemma 13.2.4 of `aeh.md`.

The natural clock here is total exponent, not blocks. The frontier is one unit of exponent per bit of start — the start's own digit supply, with no estimate in it — and it reads as $1/4$ blocks per bit only through the mean exponent per block, $\mathbb{E}_B[m+r] = 4$, the $\sigma \approx 4.0$ of Heuristic 3.9 and exactly $2+2$ under $\pi_{k,D}$. That conversion is available here precisely because the word is exactly B here. A descent from N to $O(1)$ takes $1/\beta = 1.2047\ldots$ blocks per bit, equivalently $4.8188\ldots$ units of exponent per bit, some 4.8 times as far. Hypothesis 5.1 is exactly the assertion that the law still describes the orbit after the start's digits have been spent; the digit budget locates the frontier, and this is the statistical statement on the far side of it. That frontier is the classical one: the same $\theta < 1/4$ is where Terras's count stops, and $1 - \beta/4 = \log_4 3$ is exactly the exponent Korec [3] obtains by exhausting it. It is a frontier of this technique and not of the problem — Inselmann [5] crosses it unconditionally for the trajectory envelope and the first moment, by an argument that buys the missing iteration invariance from a density notion stronger than natural density ($*$-density: every initial segment carries all but $O(N^{-\delta})$ of its mass for some $\delta > 0$, which sets of natural density one need not do) rather than from a sharper count. His horizons are in step time: he starts from the classical range of $\log_2 m$ steps of T and extends it by the factor $(1 - \log_2 \sqrt{3})^{-1} = 4.8188\ldots$. That factor is $(1/\beta)/(1/4)$ — the identity $4/\beta = 2/(2 - \log_2 3)$ — and is the same number in any time units; the endpoints read as $1/4$ and $1/\beta$ blocks per bit only after dividing by the mean exponent per block, which is a theorem where the cylinder count runs and is Hypothesis 5.1 where it does not.

AEH implies the ledger, in the form the hypothesis has: s is a coordinate of the observable, so $P(s = j) = 2^{-j}$ is a marginal of $\pi_{k,D}$ rather than a deduction, exactly at every $j < D$ and up to the cap's single tail cell above; and for every ε and every horizon rate, all but a vanishing density of the odd starting values of each size carry those frequencies along their first T_N blocks, all of them within the budget. Evaluating each odd x once, at its own dyadic scale $2^{\lfloor \log_2 x \rfloor}$, turns that array of statements into a single one: the odd integers that fail at their own scale have natural density zero (`aeh.md` Proposition 13.2.5). The union over *all* sampling scales is a different and larger object and is not controlled — the bad sets form a triangular array, each testing a horizon and a budget that move with the scale, and vanishing density in every window does not thin their unrestricted union. The error $O(2^{-k})$ of Theorem 3.8 prices a different statement — *predicting* the next exit valuation from a window — which the hypothesis does not use. The descent consequence is not stated here, because it is a theorem without the hypothesis. That all but a set of starting values of natural density zero drop below x^η, for every fixed $\eta > 0$, within $(1 - \log_2 \sqrt{3})^{-1} \log_2 x$ steps of T, hence within $O(\log x)$ blocks, is Inselmann [5, Cor. 1.4]; his [5, Thm. 1.10] is stronger than anything Hypothesis 5.1 yields here, being two-sided, uniform in the time, and unconditional, and it runs to $(\log_2 \frac{4}{3})^{-1} \log_2 m$ Syracuse steps — 4.8188 . . . times the classical range at which the cylinder count of Section 5 stops, the two measured in the same units. The first entry of the ledger — that odd steps occupy half the schedule to the same horizon — is likewise unconditional [5, Thm. 1.6], and is exactly what carries his own passage from T-time to Syracuse time. The further passage to block time needs the frequency with which a Syracuse step ends a block, a two-letter statistic of the parity word that neither theorem supplies and that $\pi_{k,D}$ does; it is therefore a consequence of Hypothesis 5.1 and not available to underwrite it. What the hypothesis supplies beyond these is distributional and is the whole of its content: the full 2^{-j} marginal at every j below the cap, the exact $\frac{1}{3}$ rate of Lemma 3.4, and the depth law, at every finite depth and past the budget at which the classical count stops. Its exceptional set is a set of *starting values* of vanishing density at a prescribed finite horizon — not a null set of orbits, and not a statement about any orbit's infinite tail. It does not exclude individual staircase tails (Remark 4.7); it does not iterate — the image of a density-one set of starts need not be density-one at the next scale, an obstruction Tao states in the same terms [4]; and by Heuristic 3.9 it cannot be reached by finite-window computation. Its content is a question about digits of 2-adic logarithms, older and broader than the Collatz problem.

Calibration. An eight-round campaign (per-orbit statistics over each cell's own conditional counts, 1,600–2,600 independent orbits per cell, under an altitude cut on the core) produced four apparent violations of $\pi_{k,D}$-uniformity, each dissolved by a sharper control: an s=3 marginal excess (artifact of small starting values); pair correlations (same); digit biases (contamination by the *bottom regime* — the fixed drainage basin of small integers, which is that cut's own complement, where deviations reach $z = 41$ but reflect particular numbers, not a measure); and finally a 5σ digit bias in a single cell surviving a 2^{40} cut. The last dissolved under fixed-horizon unweighted sampling (0.2503 ± 0.0015 against null 0.25 over 84,739 visits) and was explained exactly:

Lemma 5.2 (deterministic routing)**.** *For $\omega \equiv 1 \pmod 8$, $d = 1$: here $s = 1$, $C = 3\omega + 1$, $v_2(C) = 2$, $a_+ = 0$, so $\omega_+ = (3\omega + 1)/4$ and $d_+ = 1$; the residue $\omega \bmod 32$ determines the next class exactly: $1 \mapsto (1, 1)$ (immediate self-return), $9 \mapsto (7, 1)$, $17 \mapsto (5, 1)$, $25 \mapsto (3, 1)$, the lifting class.*

Ratio estimators over correlated visit sequences inherit this routing (residue 25 leads into deep descent, hence fewer qualifying visits, hence inflated per-orbit weight), manufacturing a bias aimed at one residue of the unique class with a self-loop channel. Uniformity stands unqualified at every tested depth and cell, within two ceilings the campaign does not reach past. It tests block lengths $L = 1$ and $L = 2$ — single cells and consecutive-pair cells, and no longer pattern than that — and is silent above; and its adjudicating runs pool per visit across orbits, so what they measure is the across-orbit average of the per-start frequencies, a consequence of Hypothesis 5.1 rather than the per-start statement itself, which would be tested by the distribution across orbits of $\|\nu - \pi_{k,D}\|_{\mathrm{TV}}$ and is not among the runs reported (`aeh.md` §13.4, §13.5). A third qualifier belongs to the campaign rather than to the hypothesis: its altitude guard cuts on the core ω_+ rather than on the door, which censors s, m_+ and a_+ at once, and it does bind at finite size — in the adjudicating run it removes 2.6% of visits and moves the reported statistics at the third decimal (`aeh.md` §13.4). Hypothesis 5.1 carries no cut of any kind; it tallies within the exponent budget, which bounds every tallied exit from below without selecting on the letter being tallied.

6 Discussion

In reduced coordinates the Collatz problem factors cleanly. The local arithmetic is under exact control: one 2-adic logarithm governs the step, and finite windows decide steps without error. What remains is a single question wearing two costumes: for typical starting values, whether the anchors' digits equidistribute (Hypothesis 5.1 — statistics); for cycles, whether closed anchor walks are rigid (Theorem 4.6 — arithmetic). The staircase shows the costumes cover the same body. Three directions have defined success criteria: rigidity statements for closed anchor walks beyond the size level (the only route past Theorem 4.6); transfer-operator analysis of the exact window chain; and any progress on digits of 2-adic logarithms of integers, a problem this paper inherits rather than creates.

A Methodology, responsibility, and records

This work was developed through extensive AI-assisted research over three years (2023–2026), from a hand-drawn origin note (13 March 2023) through seventy-nine drafts to a research wiki whose pages carry status headers, verification records, and refuted claims with their refutation data. The author is responsible for the published claims, the research direction, and the standards of evidence; the AI collaborator carried out the mathematical development within the author's coordinate system — formulations, proofs, computations, and connections to the literature. Because this division of labor is unusual, correctness claims rest on the documented verification protocol rather than on authorial authority alone: every computational claim cites a runnable script; every result labeled proved was re-verified with independently written code before being so labeled; the calibration campaign's four dissolved anomalies, and two corrected search-filter errors on the cycle side, are documented rather than smoothed over. The complete record (wiki, code, draft history, origin note) is public at https://github.com/macindoe/collatz; every wiki section and script named in this paper is cited at commit `677a76a`.

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